TOD-ProcBench: Benchmarking Complex Instruction-Following in Task-Oriented Dialogues

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Abstract

In real-world task-oriented dialogue (TOD) settings such as customer support for trip booking, banking, and healthcare, agents are required to strictly adhere to complex instructions while conducting multi-turn conversations with customers. These instructions are typically presented in natural language format and include general guidelines and step-by-step procedures with complex constraints. Existing TOD benchmarks often oversimplify the complex nature of these instructions by reducing them to simple schemas composed of intents, slots, and API call configurations. To address this gap and systematically benchmark LLMs' instruction-following capabilities, we propose **TOD-ProcBench**, a challenging ***bench**mark* featuring complex ***proc**ess* instructions with intricate, fine-grained constraints that evaluates various LLMs' abilities to understand and follow instructions in ***multi-turn TODs***. Our benchmark dataset comprises instruction documents derived from the high-quality ABCD dataset with corresponding conversations under human quality control. We formulate fine-grained constraints and action procedures as multi-level condition-action instruction statements. We design three tasks to comprehensively benchmark LLMs' complex instruction-following capabilities in multi-turn TODs. Task 1 evaluates how LLMs retrieve the most relevant statement from a complex instruction and predict the corresponding next action. In Task 2, we synthesize instruction-violating responses by injecting inconsistencies and manipulating the original instructions, and then we analyze how effectively LLMs can identify instruction-violating responses. Task 3 investigates LLMs' abilities in conditional generation of instruction-following responses based on the original complex instructions. Our benchmarking results reveal significant gaps in LLMs' instruction-following capabilities in multi-turn conversations across all three tasks. Additionally, we conduct studies on the impact of multilingual settings and different instruction text formats on compliance performance. We release our benchmark under the Llama 3.3 Community License Agreement to advance research on multi-turn TODs' complex instruction-following capabilities. The dataset is available for download at https://www.amazon.science/publications/tod-procbench-benchmarking-complex-instruction-following-in-task-oriented-dialogues.

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39th Conference on Neural Information Processing Systems (NeurIPS 2025) Workshop: Multi-Turn Interactions in Large Language Models.

1 Introduction

Task-oriented dialogue systems (ToDs) take decisive steps and actions when interacting with users with the primary goal of satisfying customer requests [Wei et al., 2018, Sekulic et al., 2024, Xu et al., 2024]. In multi-turn interactions with users, they initially comprehend a user's query and then collect information needed to address the request while following the constraints defined by complex instructions. Complex instructions comprise strict guidelines that dictate under what conditions the actions and API calls must be executed [Rastogi et al., 2020, Raimondo et al., 2023, Chen et al., 2021, Roy et al., 2024]. The complexity of these instructions varies based on the user's intent and the task domain.

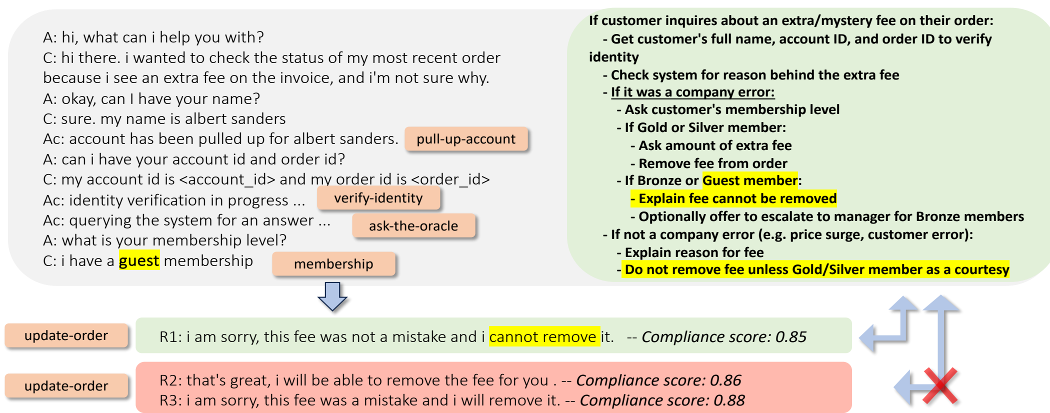

Figure 1: Workflow steps (orange boxes) from ABCD dataset show the the order of calling actions for "status_mystery_fee" intent, an inadequate representation of all nuanced constraints in the corresponding TOD-ProcBench instruction document excerpt on the right. The compliance scores are computed with ComplianceScorer.

Existing datasets either fail to capture challenges due to multi-turn conversation or fail to capture real-world instruction complexity, lacking fine-grained conditional instructions shown in Figure 1. Real-world instructions typically use natural language ***IF (Condition) - THEN (Action)*** structures which introduce ambiguity compared to symbolic formats but express more complex guidance. This creates a need for public datasets featuring multi-turn conversations that comply with fine-grained natural language conditional instructions. Figure 1 demonstrates how ComplianceScorer [Min et al., 2023] incorrectly labels all responses as compliant when responses R2 and R3 actually violate fee removal instructions, highlighting the need for fine-grained conditional instructions beyond basic workflows. Natural language instructions offer the additional benefit of seamlessly sharing consistent instructions between human agents and chatbots without conversion. For this purpose, we construct a multilingual dataset called TOD-ProcBench which contains conversations and complex instructions in the condition-action format. Our contributions are:

- We propose TOD-ProcBench, based on ABCD [Chen et al., 2021], as a collection of benchmarks with complex condition-action natural language instructions and multilingual multi-turn conversations.
- We publish benchmarks for three TOD tasks: (1) instruction and next action selection - this task measures how well LLMs can identify applicable instructions during a multi-turn conversation (2) instruction-following evaluation - this task measures how well LLMs can detect violation of complex instruction (3) instruction-following response generation - this task measures how well LLMs can respond under complex instructions written for an entire conversational interaction.
- We propose an approach for generating a test benchmark of instruction-violating responses in TOD-ProcBench by injecting different perturbations to instructions and prompting the LLM to generate responses which are similar to the ground-truth yet adhere to manipulated instructions.

- We show that all LLMs considered in this study struggle to perform accurately on (1) complex formatted instruction selection, (2) instruction-violation response evaluation, and (3) instruction-following response generation tasks. In addition, we find that LLMs have similar multilingual capability for those tasks with a subtle preference for English.

2 Related Work

In general, ToDs such as customer service chatbots, physical robots, virtual assistants [Cui et al., 2017, Sengupta et al., 2019, Lewis and White, 2023] are strictly required to comply with the predefined procedures when executing API calls and actions to resolve user issues. Datasets have been introduced to incorporate constraints on TODs through schema-guided paradigms. The schema of the synthetic SGD dataset [Rastogi et al., 2020] imposes constraints through the set of obligatory slots and the order in which they must be filled for actions to be executed. As shown in Appendix A, SGD schema for *finding next flight* intent includes slots (flight's origin and destination, both airports, seating class and etc.), their definitions and values, and required slots for each intent. Similar to SGD, STAR [Mosig et al., 2020] and FLAP [Roy et al., 2024] present the schema concept in ToDs in flowchart format. For these datasets, the schema with actions and slot values do not usually reflect the nature of real-world instructions as the latter often has more complex step-by-step structure and condition-action combination. The Action-Based Conversations Dataset (ABCD) [Chen et al., 2021] which our study is based on incorporates guidelines that a human agent needs to execute in a predefined order. We go a step further and present more realistic, natural language instructions in the form of multi-level condition-action statements that provide precise information about the conditions that are required to be true for actions to be executed. More recent datasets and studies benchmark an LLM agent's ability to follow SOPs [Nandi et al., 2025, Li et al., 2025]. Their focus is more on the accuracy of tool selection alone, complementing our study of conversational capabilities in a multi-turn dialogue setting.

3 TOD-ProcBench

3.1 Problem Overview

In our setting, to generate a response r for a given dialogue history u (for example, "user: i need to return this jacket. it is the wrong size") within a conversation in language l, a TOD system D must follow a set of conditions $\mathbf{c} \subseteq C$ and execute a sequence of corresponding actions $\mathbf{a} = \langle a_1, a_2, \ldots, a_n \rangle$ with $a_i \in A$ outlined in a given instruction $n \in N$. We present multilingual TOD-ProcBench with two main dimensions having three and seven variations, respectively: (1) the format dimension f of $n \in N$ with English natural language (f_1="Nested If-Then", f_2="Flattened If-Then", and f_3="Flattened JSON Mappings of Conditions and Actions") and (2) the language dimension l of the conversation (Arabic, Chinese, English, French, German, Hindi, Spanish).

3.2 Complex Instruction Components

3.2.1 Conditions

Specific states or attributes, such as customer attributes (e.g., membership level), item properties (e.g., purchase date), or conversation states (e.g., user intent) must be verified before executing a sequence of corresponding actions.

3.2.2 Actions

The TOD must execute specific operations or responses when associated conditions are satisfied. Actions are organized hierarchically with conditions. For example, when a user inquires about the return policy (1st-level condition), the agent must first identify the user's membership level (1st-level action). If the user has a *silver* membership (2nd-level condition), then the agent provides specific information like "only purchases in the last 6 months are returnable" (2nd-level action).